

ADIDAS — Retail Customer Retention Analytics

Power BI Project Report

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TASK 1 — Data Modelling & Cleaning

- Five datasets were loaded into Power BI
- No duplicates found in any dataset.
- Missing values — 353 nulls in Churn Reason filled with "Active" .
- Data types verified — dates set to Date format, Amount to Decimal Number

Calculated Columns Created

```
Membership_Duration_Days=DATEDIFF(Customer_Demographics[Membership_Since],TODAY(),DAY)

Membership_Duration_Years=DATEDIFF(Customer_Demographics[Membership_Since],TODAY(),DAY)/365.25

Age_Group = IF(Customer_Demographics[Age]<=24,"18-24",IF(Customer_Demographics[Age]<=34,"25-34",IF(Customer_Demographics[Age]<=44,"35-44",IF(Customer_Demographics[Age]<=45,"45-54","55+"))))

Transaction_Year = YEAR(Customer_Transactions[Transaction_Date])

Transaction_Month = MONTH(Customer_Transactions[Transaction_Date])

CLV = DIVIDE(CALCULATE(SUM(Customer_Transactions[Amount])),RELATEDTABLE(Customer_Transactions)),DATEDIFF(Customer_Demographics[Membership_Since],DATE(2025,9,1),DAY) / 365.25)

CLV_Segment = IF(Customer_Demographics[CLV] <= 131.52, "Low CLV",IF(Customer_Demographics[CLV] <= 279.18, "Medium CLV","High CLV"))

Purchase_Segment = SWITCH(TRUE(),Customer_Demographics[Transaction count]=0,"No Purchase",Customer_Demographics[Transaction count]<=3,"Low-Tier(0-3)",Customer_Demographics[Transaction count]<=8,"Mid-Tier(4-8)","High-Tier(+9)")
```

Data Model — Snowflake Schema

Churn ■■■■■■■■■■■■■■■■■■■■■■→ Customer_Demographics ■■■→ Customer_Transactions ←■■■ Store_Locations

Loyalty_Program ■■■■■→ Customer_Demographics

TASK 2 — Churn & Retention Metrics

1.Churn Rate = (Churned Customers / Total Customers) × 100

Key Findings:

Region Churn Rate Finding

Asia-Pacific 33.33% Highest — needs urgent attention

Middle East 28.99% Above average

North America 24.49% Below average

South America 24.24% Below average

Europe 24.00% Lowest — best retention

2. Churn by Income Level

Channel Churn Rate Finding

Online 28.95% Higher churn

Store 25.68% Better retention

3. Churn by Loyalty Tier

Tier Churn Rate Alert

Elite 44.83% CRITICAL

Premium 32.26% HIGH

Plus 25.00% MEDIUM

Base 22.48% LOW

Customer Funnel

Total Customers → 300 (100%)

Repeat Customers → 256 (85.3%)

Churned Customers → 82 (27.3%)

TASK 3 — Repeat Purchase Analysis

Purchase Segmentation:

Key Findings:

Average Purchase Frequency by Region

Average Purchase Frequency by Age Group

Average Purchase Frequency by Loyalty Tier

Most Purchased Categories

TASK 4 — Promotion & Loyalty Impact

Promotion Analysis:

Total Transactions → 1,000

With Promotion → 527 (52.7%)

Without Promotion → 473 (47.3%)

Avg Spend WITH Promotion → \$272

Avg Spend WITHOUT Promotion → \$259

Difference → +\$13 (+5.27%)

Points Analysis by Loyalty Tier

Key Insights

Promotions drive 5.27% higher average spend

41% of earned points go unredeemed

Elite tier has highest churn (44.83%) despite high redemption

Plus tier earns most points but redeems least

Recommendations

- Fix Elite tier — 44.83% churn is critical
- Improve points redemption — 41% of points unused
- Make promotions tier-specific not mass discounts

TASK 5 — Store & Channel Performance

Store Performance:

Store Opening Year vs Retention

- Best retention: 2022 stores (81.82%) and 2012 stores (81.58%)
- Worst retention: 2017 stores (56.25%)
- Newer stores generally have lower retention as they haven't built loyal base yet.

Key Insights

- Franchise stores generate highest revenue (35.7% of transactions)
- Flagship stores are underutilised (only 16.6% of transactions)
- Retention rate varies significantly by store opening year

TASK 6 — Customer Lifetime Value Analysis

Key Insights

- Middle East has highest CLV (\$909) — prioritize retention
- Asia-Pacific has lowest CLV (\$296) AND highest churn — double problem
- Elite tier has lowest CLV despite being top tier — paradox!

Task 7: Final Dashboard & Executive Summary

TOP 3 STRATEGIC RECOMMENDATIONS FOR ADIDAS

- Completely redesign Elite benefits — add free shipping always, exclusive product launches, VIP event access, dedicated customer service hotline
- Partner with premium lifestyle brands for cross-promotional offers (luxury fitness clubs, travel brands, etc.) and
- Add AR Virtual Try-On for footwear and apparel — this reduces return rates and increases purchase confidence significantly

Segment	Customers	Customers	%	%
Low-Tier (0-3 purchases)	Low-Tier (0-3 purchases)	152	152	50.67
Mid-Tier (4-8 purchases)	Mid-Tier (4-8 purchases)	128	128	42.67

Segment	Customers	Customers	%	%
High-Tier (9+ purchases)	High-Tier (9+ purchases)	2	2	0.67
No Purchase	No Purchase	18	18	6

Region	Avg Transactions
Europe	3.60
Asia-Pacific	3.59
Middle East	3.57
South America	3.56
North America	3.38

Age Group	Avg Transactions
18-24	3.91
35-44	3.82
25-34	3.44
45-54	3.38
55+	3.37

Loyalty Tier	Avg Transactions
Elite	3.77
Base	3.67
Premium	3.58
Plus	3.25

Category	All Customers	Loyal Customers
Apparel	34.1%	31.39%
Footwear	33.2%	33.98%
Accessories	32.7%	34.63%

Tier	Avg Earned	Avg Redeemed	Redemption Rate
Elite	4,471	3,347	74.87%
Base	4,842	2,805	57.92%
Premium	5,317	3,056	57.48%
Plus	5,587	3,038	54.37%

Store Type	Avg Transaction	Revenue	Churn Rate
Franchise	\$271	\$96,913	26.9%

Store Type	Avg Transaction	Revenue	Churn Rate
Outlet	\$269	\$61,932	27.1%
Online	\$260	\$64,282	26.8%
Flagship	\$258	\$42,850	31.3%